

Applied Linear Algebra (EE 635): Problem Set 4

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August 22, 2025

Questions

1. Let $x_1, \dots, x_n$ be fixed numbers. The matrix below, called a *Vandermonde matrix*, occurs in applications such as signal processing, error-correcting codes, and polynomial interpolation.

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{bmatrix}.$$

Given $y = (y_1, \dots, y_n) \in \mathbb{R}^n$, suppose $c = (c_0, \dots, c_{n-1}) \in \mathbb{R}^n$ satisfies $Vc = y$, and define the polynomial

$$p(t) = c_0 + c_1 t + c_2 t^2 + \cdots + c_{n-1} t^{n-1}.$$

- (a) Show that $p(x_1) = y_1, \dots, p(x_n) = y_n$. We call $p(t)$ an interpolating polynomial for the points $(x_1, y_1), \dots, (x_n, y_n)$ because the graph of $p(t)$ passes through the points.
 - (b) Suppose $x_1, \dots, x_n$ are distinct numbers. Show that the columns of V are linearly independent. [Hint: How many zeros can a polynomial of degree $n - 1$ have?]
 - (c) Prove: “If $x_1, \dots, x_n$ are distinct numbers, and $y_1, \dots, y_n$ are arbitrary numbers, then there is an interpolating polynomial of degree $\leq n - 1$ for $(x_1, y_1), \dots, (x_n, y_n)$.”
2. Find a matrix A such that the transformation $x \mapsto Ax$ maps $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ into $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$, respectively.
[Hint: Write a matrix equation involving A , and solve for A .]
 3. Let A be a 6×4 matrix and B a 4×6 matrix. Show that the 6×6 matrix AB cannot be invertible.

4. Let V be a subspace of $\mathbb{R}^n$ and let $v_1, v_2, \dots, v_k \in V$.

- (a) Prove that $\text{span}\{v_1, v_2, \dots, v_k\}$ is a subspace of V .
- (b) Prove that

$$\text{span}\{v_1, v_2, \dots, v_k\} = \bigcap \{W \subseteq \mathbb{R}^n : W \text{ is a subspace and } v_1, v_2, \dots, v_k \in W\}.$$

In Questions 5 and 6, A denotes an $m \times n$ matrix. Mark each statement **True** or **False**. Justify each answer.

- 5.
 - (a) The null space of A is the solution set of the equation $Ax = 0$.
 - (b) The null space of an $m \times n$ matrix is in $\mathbb{R}^m$.
 - (c) The column space of A is the range of the mapping $x \mapsto Ax$.
 - (d) If the equation $Ax = b$ is consistent, then $\text{Col } A$ is $\mathbb{R}^m$.
 - (e) The kernel of a linear transformation is a vector space.
 - (f) $\text{Col } A$ is the set of all vectors that can be written as Ax for some x .
- 6.
 - (a) A null space is a vector space.
 - (b) The column space of an $m \times n$ matrix is in $\mathbb{R}^m$.
 - (c) $\text{Col } A$ is the set of all solutions of $Ax = b$.
 - (d) $\text{Nul } A$ is the kernel of the mapping $x \mapsto Ax$.
 - (e) The range of a linear transformation is a vector space.
 - (f) The set of all solutions of a homogeneous linear differential equation is the kernel of a linear transformation.